

Load Dataset

```
In [1]: import pandas as pd

dataset = pd.read_csv("spam.csv", encoding='latin1')

data = dataset[['v1', 'v2']].copy()
data.columns = ['label', 'message']

print("Total messages:", data.shape[0])

Total messages: 5572
```

```
In [3]: print("\nSample data:")
dataset
```

Sample data:

Out[3]:

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN
3	ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
4	ham	Nah I don't think he goes to usf, he lives aro...	NaN	NaN	NaN
...	...	...	...	...	...
5567	spam	This is the 2nd time we have tried 2 contact u...	NaN	NaN	NaN
5568	ham	Will i_ b going to esplanade fr home?	NaN	NaN	NaN
5569	ham	Pity, * was in mood for that. So...any other s...	NaN	NaN	NaN
5570	ham	The guy did some bitching but I acted like i'd...	NaN	NaN	NaN
5571	ham	Rofl. Its true to its name	NaN	NaN	NaN

5572 rows × 5 columns

Data Preprocessing

```
In [4]: # Label encoding
data['label_num'] = data['label'].map({'ham': 0, 'spam': 1})

# Check label distribution
print("\nLabel distribution:")
print(data['label'].value_counts())
```

```
Label distribution:
label
ham      4825
spam      747
Name: count, dtype: int64
```

Train-Test Split

```
In [5]: from sklearn.model_selection import train_test_split
        from sklearn.feature_extraction.text import TfidfVectorizer

        # Train-test split
        X = data['message']
        y = data['label_num']

        X_train, X_test, y_train, y_test = train_test_split(
            X, y, test_size=0.2, random_state=42, stratify=y)

        print("\nTrain size:", X_train.shape[0])
        print("Test size:", X_test.shape[0])

        # TF-IDF Vectorizer - text to numeric feature conversion
        vectorizer = TfidfVectorizer(stop_words='english')

        # Fit on train data and transform
        X_train_vec = vectorizer.fit_transform(X_train)

        # Transform test data
        X_test_vec = vectorizer.transform(X_test)

        print("\nSample features shape (train):", X_train_vec.shape)
```

Train size: 4457

Test size: 1115

Sample features shape (train): (4457, 7440)

Model Training and Evaluation

```
In [6]: from sklearn.linear_model import LogisticRegression
        from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

        # Step 4: Model training
        model = LogisticRegression(max_iter=1000) # max_iter badhaya for convergence
        model.fit(X_train_vec, y_train)

        # Predict on test set
        y_pred = model.predict(X_test_vec)

        # Accuracy
        acc = accuracy_score(y_test, y_pred)
        print(f"\nTest Accuracy: {acc*100:.2f}%")

        # Detailed classification report
        print("\nClassification Report:")
        print(classification_report(y_test, y_pred, target_names=['Ham', 'Spam']))

        # Confusion matrix
        cm = confusion_matrix(y_test, y_pred)
        print("\nConfusion Matrix:")
        print(cm)
```

Test Accuracy: 96.59%

Classification Report:

	precision	recall	f1-score	support
Ham	0.96	1.00	0.98	965
Spam	0.99	0.75	0.86	150
accuracy			0.97	1115
macro avg	0.98	0.88	0.92	1115
weighted avg	0.97	0.97	0.96	1115

Confusion Matrix:

```
[[964  1]
 [ 37 113]]
```

Predict on Full Dataset and Save Results

```
In [7]: # Predict on entire dataset
X_all_vec = vectorizer.transform(data['message']) # vectorize full messages
data['predicted_label_num'] = model.predict(X_all_vec) # predict spam/ham numeric
data['predicted_label'] = data['predicted_label_num'].map({0: 'ham', 1: 'spam'})
# Show sample results (actual vs predicted)
print("\nSample predictions:")
print(data[['message', 'label', 'predicted_label']].head(10))

# Save to new CSV file
data.to_csv("spam_predictions.csv", index=False)
print("\nPredictions saved in 'spam_predictions.csv'")
```

Sample predictions:

	message	label	predicted_label
0	Go until jurong point, crazy.. Available only ...	ham	ham
1	Ok lar... Joking wif u oni...	ham	ham
2	Free entry in 2 a wkly comp to win FA Cup fina...	spam	spam
3	U dun say so early hor... U c already then say...	ham	ham
4	Nah I don't think he goes to usf, he lives aro...	ham	ham
5	FreeMsg Hey there darling it's been 3 week's n...	spam	ham
6	Even my brother is not like to speak with me. ...	ham	ham
7	As per your request 'Melle Melle (Oru Minnamin...	ham	ham
8	WINNER!! As a valued network customer you have...	spam	spam
9	Had your mobile 11 months or more? U R entitle...	spam	spam

Predictions saved in 'spam_predictions.csv'

```
In [8]: user_input = input("Enter 0 to mark all messages as SPAM, 1 to mark all as HAM, or press Enter to use model prediction: ")

if user_input == '0':
    # spam
    data['final_label'] = 'spam'
    print("\nAll messages marked as SPAM:")
    print(data[['message', 'final_label']].head(10))

elif user_input == '1':
    # ham
    data['final_label'] = 'ham'
    print("\nAll messages marked as HAM:")
    print(data[['message', 'final_label']].head(10))

else:
    # Model prediction
    data['final_label'] = data['predicted_label']
    print("\nModel predictions:")
    print(data[['message', 'final_label']].head(10))
```

Enter 0 to mark all messages as SPAM, 1 to mark all as HAM, or press Enter to use model prediction: 1

All messages marked as HAM:

	message	final_label
0	Go until jurong point, crazy.. Available only ...	ham
1	Ok lar... Joking wif u oni...	ham
2	Free entry in 2 a wkly comp to win FA Cup fina...	ham
3	U dun say so early hor... U c already then say...	ham
4	Nah I don't think he goes to usf, he lives aro...	ham
5	FreeMsg Hey there darling it's been 3 week's n...	ham
6	Even my brother is not like to speak with me. ...	ham
7	As per your request 'Melle Melle (Oru Minnamin...	ham
8	WINNER!! As a valued network customer you have...	ham
9	Had your mobile 11 months or more? U R entitle...	ham

Enter 0 to mark all messages as SPAM, 1 to mark all as HAM, or press Enter to use model prediction: 0

All messages marked as SPAM:

	message	final_label
0	Go until jurong point, crazy.. Available only ...	spam
1	Ok lar... Joking wif u oni...	spam
2	Free entry in 2 a wkly comp to win FA Cup fina...	spam
3	U dun say so early hor... U c already then say...	spam
4	Nah I don't think he goes to usf, he lives aro...	spam
5	FreeMsg Hey there darling it's been 3 week's n...	spam
6	Even my brother is not like to speak with me. ...	spam
7	As per your request 'Melle Melle (Oru Minnamin...	spam
8	WINNER!! As a valued network customer you have...	spam
9	Had your mobile 11 months or more? U R entitle...	spam